

Bridging linguistic insight and NLP Innovation for French-based Creoles

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Rationale

1. Creole NLP is underdeveloped
2. Creole languages are low-resource
3. Dominant-language bias and overgeneralization masks Creole typological complexity
4. Leveraging Creole-to-Creole relationships for NLP

Martinican and Haitian creoles

- Both are French-lexified.
- SVO Creoles, morphologically reduced: no inflection for gender or number.
- Tense, Mood, Aspect (TMA) expressed with separate morphemes.
- Close syntactic structure; lexical overlap but with notable variation.

*The mom lik**ed** her son**s** – La maman aim**ait** ses fil**s***
[MC] Manman-lan **té** enmen **sé** tchanmay-li **a**
[HC] Manman an **te** renmen pitit li **yo**



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Objectives

1. Advance NLP for Creole languages
2. Understand linguistic challenges
3. Go beyond aggregate evaluation
4. Guide resource development
5. Build towards real-world applications

Recommendations for Overcoming Linguistic Barriers in Healthcare: Challenges and Innovations in NLP for Haitian Creole (Mompelat, 2025)

- Spoken by millions and yet remains underrepresented in NLP.
- Language barriers contribute to miscommunication, misdiagnoses, and treatment disparities.
- Proposal : linguistically-informed NLP approach to a culturally-accurate AI-assisted medical translation system
 - In collaboration with Dr. David Tézil (University of Alabama) and Rose Accilien (University of Miami)

Recommendations for Overcoming Linguistic Barriers in Healthcare: Challenges and Innovations in NLP for Haitian Creole (Mompelat, 2025)

Motivated (and challenged) by

1. High variation (*li mouri deu kansèr vs. se kansè ki touye l*)
2. Frequent code-switching/code-mixing
3. Specific medical terminology (*pwatrin vs. lestomak, dyabèt vs. maladi sik*)
4. Low-resource context
5. Typological similarity with other Creoles but differences with lexifier language

Roadmap

End Goal: Haitian Creole Medical MT Pipeline

Reverse journey: From MT back to LID

- **Stage 1:** Parsing (HC $\leftrightarrow$ MC transfer)
- **Stage 2:** POS tagging & cascading errors
- **Stage 3:** Language Identification (LID)

Current vision: Scaffolded modeling vs. end-to-end pipeline

Wait... What is Parsing?

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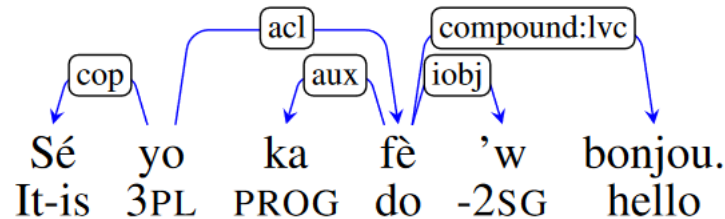


Stage 1: Parsing
(HC ↔ MC transfer)

- What is syntactic Parsing ? Analyzing **syntactic structure** of a sentence.
- Two layers of annotations :
 - **POS tagging** → lexical category of a word
 - Dependency parsing → grammatical role in clause
- Enables: **Machine Translation, Text Prediction, Grammar Tools**, etc.

“Sé yo ka fè’w bonjou” (They are the ones greeting you)

- Treebank:



Treebank efforts in the community :

Universal Dependencies (Nivre et al. 2020) → Standardized annotation framework used in **240+ treebanks** and **100+ languages**.

Token	POS
Sé	AUX
yo	PRON
ka	PART
fè	VERB
'	PUNCT
w	PRON
bonjou	NOUN
.	PUNCT



The challenge : low-resource languages

- Creoles = **low-resource** → small, domain-limited datasets
- Language Models require **large amounts of annotated data**
- Strategies to maximize data:
 - Cross-lingual transfer (**French, HC, or MC**)
 - Training techniques : Concatenation, Transfer Learning or Multitask Learning strategies
 - Combine and jointly train small datasets

Advancements in Creole NLP: Overcoming Data Scarcity in Creole Syntactic Parsing (Mompelat 2025)

1. Develop robust dependency parsers for *Martinican Creole* (Mompelat 2022) and *Haitian Creole*
2. Evaluate and compare data concatenation and multi-task learning (MTL) strategies.
3. Identify pathways to improve parsing accuracy in low-resource settings and guide future annotation work.

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Stage 1: Parsing
(HC ↔ MC transfer)

Experimental Setup

- **Treebanks used:**
 - UD French (Guillaume et al. 2019) → high-resource reference
 - Martinican Creole (MC) (Mompelat) → 236 sentences
 - Haitian Creole (HC) (Jagodzińska et al.) → 144 sentences
- **Training designs:**
 - **Concatenation:** Merge HC + MC
 - **Multi-task learning (MTL):** Train jointly with task-specific components
- **Models tested:**
 - CamemBERT v1 & v2
 - RoBERTa-base
 - CreoleVal (Creole-specific fine-tuning)

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Stage 1: Parsing
(HC ↔ MC transfer)



Key Findings

Creole-to-Creole transfer > French-to-Creole (in some instances)

→ MC & HC can support each other better than French alone

- CreoleVal = most robust for rare function words
- Model fusion strategies show promise for low-resource contexts
- Practical takeaway: Cross-Creole collaboration can be more effective than relying on dominant languages



Next Steps

- Expand Creole treebanks with targeted annotation
- Implement **model fusion strategies for parsers**
- Support pipeline for healthcare translation systems and more
- Parsing errors traced to POS tagging confusions
→ motivates deeper POS tagging study

Simple Morphology, Complex Models: A Benchmark Study and Error Analysis of POS Tagging for Martinican Creole (Mompelat 2025)

POS tagging = foundation for NLP pipelines

- In low-resource settings, tagging errors can cascade upward → Parsing, MT
- MC = high structural ambiguity → polyfunctional tokens
- Treat POS tagging as primary task, not byproduct of parsing

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Stage 2: POS
tagging

Martinican creole

- **French-lexified Creole:** heavy lexical overlap with French, but isolating morphology (no agreement, minimal inflection).
- **Transparent orthography:** unlike French's diverging orthography, spelling in MC closely matches pronunciation.
- **Code-switching & contact:** frequent co-occurrence with French across domains makes MC ideal for testing multilingual and low-resource models.
- **Ambiguity from polyfunctionality:** many short tokens fulfill multiple syntactic roles without overt marking.

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Stage 2: POS tagging

KON = CCONJ, SCONJ, ADP

- a. Rad-maré anni balié lakot Atlantik
tidal-wave only sweep coast atlantic
kon lakot karayib
as_well_as coast caribbean
'The tidal wave only swept the Atlantic
Coast **as well as** the Caribbean Coast'
CCONJ
- b. Kon di Kolo, 'si ou pa ri yo,
like say Kolo, 'if 2SG not laugh 3PL,
yo ké ri'w'.
3PL FUT laugh'2PL
'**Like** Kolo says, "If you don't laugh at
them, they'll laugh at you" SCONJ
- c. Jik ki tan nou ké asepte yo
until which time we FUT accept 3PL
fè nou wont **kon** sa ?
make 2PL shame like that
'Until when will we let them embarrass
us like that?' ADP

KI = PRON, SCONJ, DET

- a. Sé pa mwen **ki** pou réparé
COP not me who must fix
lektrisité [...]
electricity
'I'm not the one **who** must fix the elec-
tricity [...]' PRON
- b. chak moun-an ja ka di **ki**
each person-DET already PROG say that
yo pa dakò.
3PL not agree
'Each of them are already saying **that**
they disagree' SCONJ
- c. Jik **ki** tan nou ké asepte yo
until which time we FUT accept 3PL
fè nou wont kon sa ?
make 2PL shame like that
'Until when will we let them embarrass
us like that?' DET



Dataset & Models

MC POS corpus

- 534 sentences (9,470 tokens) from news, blogs, social media.

Models tested

- mBERT – broad multilingual baseline.
- XLM-Roberta – stronger multilingual pretraining.
- CreoleVal – domain-adapted XLM-R model for Creoles, no MC data.

Results

Overall performance

- mBERT: 92%
- XLM-R: 91%
- CreoleVal: 94% (best overall F1)

Weak categories

- Low-frequency or ambiguous: INTJ, SCONJ, ADP, ADJ/VERB/NOUN overlaps.
- Polyfunctional tokens (*ki*, *kon*, *pou*) remain persistent error sources.

Complementary strengths identified between models

- CreoleVal: excels on function words, lowest OOV error rate (13%).
- mBERT: stronger on conjunctions/prepositions.
- XLM-R: best on NUM, rare morphosyntactic items.

Implication

- CreoleVal overall “best” model; but complementary error profiles motivate fusion and targeted annotation.



Error Analysis – Cascading Error Profile

Aggregate scores hide variation

- Models perform well overall but fail systematically on certain categories. CreoleVal performed best overall.

Polyfunctionality-driven errors

- *ki* (PRON / SCONJ / DET): error rates up to **17%**.
- *kon* (CCONJ / SCONJ / ADP): error rates up to **67%**.
- *pou* (ADP / SCONJ): ~20% misclassified across models.
- *sé* (DET / COP): CreoleVal resolves distinction better than others.

Error Analysis – Cascading Error Profile

Confusion patterns

- Frequent swaps: **ADJ ↔ VERB ↔ NOUN** (content words).
 - Reflects the predicative nature of ADJ in Creoles
“Répons-lan klè” the answer is clear
 - Consequence of lack of inflectional morphology on VERB
Alé-vini (VERB = to come and go / NOUN = a back and forth)
Alié (VERB = to form an alliance / NOUN = an ally)
- Functional overlap: **ADP ↔ SCONJ ↔ CCONJ**
Pou (in order to, must, so that), kon (like, as, as well as), jik (until), san (without)



OOV & Transition to LID

OOV tokens = 1.5% (144 unseen tokens)

- CreoleVal lowest OOV error rate → 13.2%
 - followed by XLM-R (16%) and mBERT (20%).

French-like forms frequently misclassified:

- *Lavé, payé, prononcé* → ADJ/VERB confusion
- *diféran* → NOUN/ADJ confusion

Loanwords & phonological variants often mislabeled as NOUN instead of X (*Galaxie, gravitation*)

- Why this matters:
 - Many errors stem from code-switching, borrowing, and overlap with French → need explicit language identification step

Compact Quantum Circuits for Martinican Creole LID: A Typologically Informed Proof-of-Concept (Mompelat 2025)

- **Goal:** Detect whether each token is Creole or French/other language
 - Seeks to handle **code-switching** in real-world contexts (e.g., Martinique, Miami)
- **Problem:** Overlap between Creole and French → hard to separate
- **Proof-of-concept:**
 - Exploration of quantum-inspired models with compact, interpretable features (entropy, vowel ratio, syllable complexity)
 - Foundation for **clean inputs** to POS → Parsing → MT
 - Entanglement allows higher-order representations. Makes it a viable area of exploration.



Current vision: Scaffolded modeling

Scaffolding NLP processes

- Treat core components (POS tagging, dependency parsing, MT) as **modular layers**.
- Optimize each individually
- Merge each modules (via scaffolding or joint-training/MTL)

Fusion strategies for POS and Parsing (planned)

- **Weighted voting:** per-tag weights based on validation F1.
- **Per-tag delegation:** assign each tag to its best-performing model (e.g. CreoleVal for function words, mBERT for connectors, XLM-R for numerals).
- **Multi-task fusion:** combine embeddings into a unified classifier.



Putting the Linguistics in Computational Linguistics

Targeted annotation (next step)

- Focus on *polyfunctional tokens* (ki, kon, pou, sé).
- Expand under-represented tags (INTJ, NUM, SCONJ, ADP, PROPN).

Danger: tailoring models too narrowly to idiosyncrasies of the dataset.

- **Linguistic alignment** = Anchor error analysis in **typology** of Martinican Creole.

Ambiguity vs. noise

- **True ambiguity** (e.g., *ki* = PRON/SCONJ/DET): more data = better disambiguation.
- **Overlapping French–Creole tokens** (e.g., loanwords, cognates): more data risks reinforcing confusion → need careful cross-lingual checks + Domain expansion



Linguistics x Computational

Two sides of improvement

- *Computational*: model fusion, ensembles, error-driven architectures.
- *Linguistic*: error analysis, typological grounding, avoiding overfitting.

Low-resource setting

- Scarcity of data makes “brute force” training impossible.
- Every annotation decision must be **strategic** and linguistically motivated.
- Computational strategies must be paired with linguistically guided annotation.

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Thank you 😊

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